Exploring Cell-Type Shifts Between Healthy and COPD Lung Tissue Using CELLxGENE Discover

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Disclaimer: Generative AI was used to assist with drafting report information and templating and writing code in this assignment; however, this writing was checked manually and edited by hand to ensure accuracy.

Source: OpenAI. (2026). *ChatGPT (GPT-5.4 Thinking)* [Large language model]. https://chatgpt.com/.

Introduction

Chronic obstructive pulmonary disease (COPD) is a lung condition caused by damage to the lungs, which leads to swelling and irritation (a.k.a. inflammation), limited airflow to the lungs, and trouble breathing (Mayo Clinic). It is most frequently caused by long-term exposure to irritating smoke or fumes, especially cigarette smoke. The two most common types of COPD are chronic bronchitis, when the lining of the tubes that bring air into the lungs called bronchi are inflamed, and emphysema, which damages the small air sacs of the lungs called alveoli and prevent them from passing enough oxygen to the bloodstream (Mayo Clinic). To understand how COPD develops, an important type of investigation is to find the types of cells that are causing the condition.

There are many distinct cell populations contained in the lung, including epithetial (outer surface) cells. The alveolar type 1 (AT1) cells and alveolar type 2 cells are particularly significant for maintaining gas exchange in the alveolar environment (Schiller et al). A previous study published in the *American Journal of Respiratory Cell and Molecular Biology* has explored the contents of lung cells with an isolated analysis technique known as single-cell RNA sequencing, which revealed a significant presence of AT2 cells in the lungs of those with COPD (Watanave et al). The dataset used in the study has been shared publicly, allowing further investigation into the provided single-cell RNA-seq data. This project aims to elaborate on this analysis from a visualization perspective by exploring the following question: **How do cell-type compositions and selected marker gene expression patterns differ between healthy and COPD lung tissue?**

Data Preparation

The data used in this project comes from the study “Anomalous Epithelial Variations and Ectopic Inflammatory Response in Chronic Obstructive Pulmonary Disease” and is made publicly available by the study authors through CZ CELLxGENE Discover, a data platform developed by the Chan Zuckerberg Initiative. The dataset was downloaded in H5AD format, which is a file type commonly used for single-cell genomics data. This format stores the data as an AnnData object, allowing both cell-level metadata and gene expression measurements to be accessed in a structured way for analysis in R.

To prepare the data, the H5AD file was read into R using the anndata package with support from reticulate. The cell-level metadata were extracted from the "obs" component of the AnnData object and converted into a data frame. From this metadata, a smaller set of variables relevant to the research question was retained, including sample identifiers, donor identifiers, disease status, cell type, sex, tissue, sample collection site, tissue type, and sample source. In addition, expression values were extracted for a small set of selected genes of interest: CXCL1, CXCL8, SFTPC, and AGER. These genes were chosen because they are biologically relevant to lung epithelial function and inflammatory signaling in COPD. The filtered gene expression data were then combined with the selected metadata variables to create a cleaned analysis dataset for exploratory analysis and visualization.

```
ad <- read_h5ad("data.h5ad")

cols_keep <- c(
  "sample_id",
  "donor_id",
  "disease",
  "cell_type",
  "sex",
  "tissue",
  "sample_collection_site",
  "tissue_type",
  "sample_source"
)

metadata_df <- as.data.frame(ad$obs[, cols_keep])
gene_names <- as.character(ad$var_names)

genes_keep <- c(
  CXCL1 = "ENSG00000163739",
  CXCL8 = "ENSG00000169429",
  SFTPC = "ENSG00000168484",
  AGER  = "ENSG00000204305"
)

genes_present <- genes_keep[genes_keep %in% gene_names]

expr_df <- as.data.frame(ad[, unname(genes_present)]$to_df())
colnames(expr_df) <- names(genes_present)

full_df <- cbind(metadata_df, expr_df)
```

Dataset characteristics

The following code computes the overall size of the analytic dataset, including the numbers of cells, variables, donors, samples, disease groups, and cell types.

```
n_rows <- nrow(full_df)
n_cols <- ncol(full_df)
n_donors <- n_distinct(full_df$donor_id)
n_samples <- n_distinct(full_df$sample_id)
n_diseases <- n_distinct(full_df$disease)
n_cell_types <- n_distinct(full_df$cell_type)

data_summary <- data.frame(
  Metric = c(
    "Number of cells",
    "Number of variables",
    "Number of donors",
    "Number of samples",
    "Number of disease groups",
    "Number of cell types"
  ),
  Value = c(n_rows, n_cols, n_donors, n_samples, n_diseases, n_cell_types)
)

kable_style(data_summary, caption = "Dataset characteristics")
```

Dataset characteristics

Metric	Value
Number of cells	57918
Number of variables	13
Number of donors	16
Number of samples	16
Number of disease groups	2
Number of cell types	21

Variables included

The cleaned dataset includes donor- and sample-level identifiers, disease status, cell type, sex, tissue descriptors, sample source information, and the four selected gene expression variables.

The following code lists the variables retained in the cleaned dataset and their corresponding data types.

```
variable_summary <- data.frame(
  Variable = colnames(full_df),
  Type = sapply(full_df, class)
)

kable_style(variable_summary, caption = "Variables included in the cleaned dataset")
```

Variables included in the cleaned dataset

	Variable	Type
sample_id	sample_id	factor
donor_id	donor_id	factor
disease	disease	factor
cell_type	cell_type	factor
sex	sex	factor
tissue	tissue	factor
sample_collection_site	sample_collection_site	factor
tissue_type	tissue_type	factor
sample_source	sample_source	factor
CXCL1	CXCL1	numeric
CXCL8	CXCL8	numeric
SFTPC	SFTPC	numeric
AGER	AGER	numeric

Missing data

Missingness is evaluated at the variable level so that any incomplete metadata or gene expression fields can be identified before interpreting results.

The following code summarizes the amount of missing data for each variable so that completeness can be assessed systematically.

```
missing_summary <- data.frame(
  Variable = colnames(full_df),
  Missing_Count = sapply(full_df, function(x) sum(is.na(x))),
  Missing_Proportion = sapply(full_df, function(x) mean(is.na(x)))
)

kable_style(missing_summary, caption = "Missingness summary")
```

Missingness summary

	Variable	Missing_Count	Missing_Proportion
sample_id	sample_id	0	0
donor_id	donor_id	0	0
disease	disease	0	0
cell_type	cell_type	0	0

	Variable	Missing_Count	Missing_Proportion
sex	sex	0	0
tissue	tissue	0	0
sample_collection_site	sample_collection_site	0	0
tissue_type	tissue_type	0	0
sample_source	sample_source	0	0
CXCL1	CXCL1	0	0
CXCL8	CXCL8	0	0
SFTPC	SFTPC	0	0
AGER	AGER	0	0

Distribution of key categorical variables

The following code tabulates the number of cells observed in each disease group.

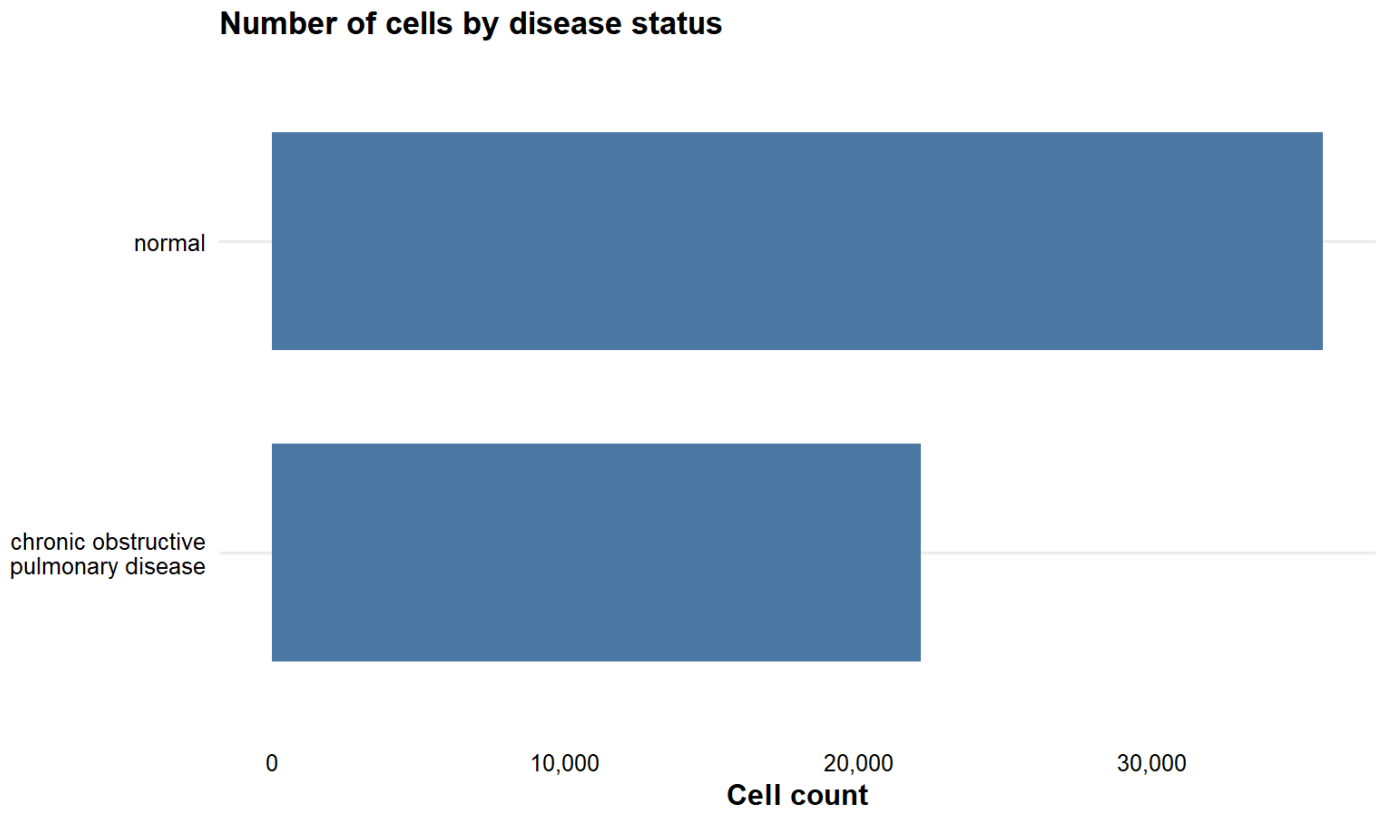
```
disease_counts <- full_df %>%
  count(disease, sort = TRUE) %>%
  mutate(disease = fct_reorder(disease, n))

kable_style(disease_counts, caption = "Cell counts by disease status")
```

Cell counts by disease status

disease	n
normal	35799
chronic obstructive pulmonary disease	22119

The following figure visualizes the distribution of cells across disease groups.

```
ggplot(disease_counts, aes(x = disease, y = n)) +
  geom_col(width = 0.7, fill = "#4C78A8") +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  scale_x_discrete(labels = wrap_axis) +
  labs(
    title = "Number of cells by disease status",
    x = NULL,
    y = "Cell count"
  )
```



Distribution of cells across disease groups. Bars are ordered from smallest to largest group size to make differences easier to compare.

The following code tabulates the most common cell types in the analytic dataset.

```
cell_type_counts <- full_df %>%
  count(cell_type, sort = TRUE)

kable_style(head(cell_type_counts, 15), caption = "Top 15 cell types by count")
```

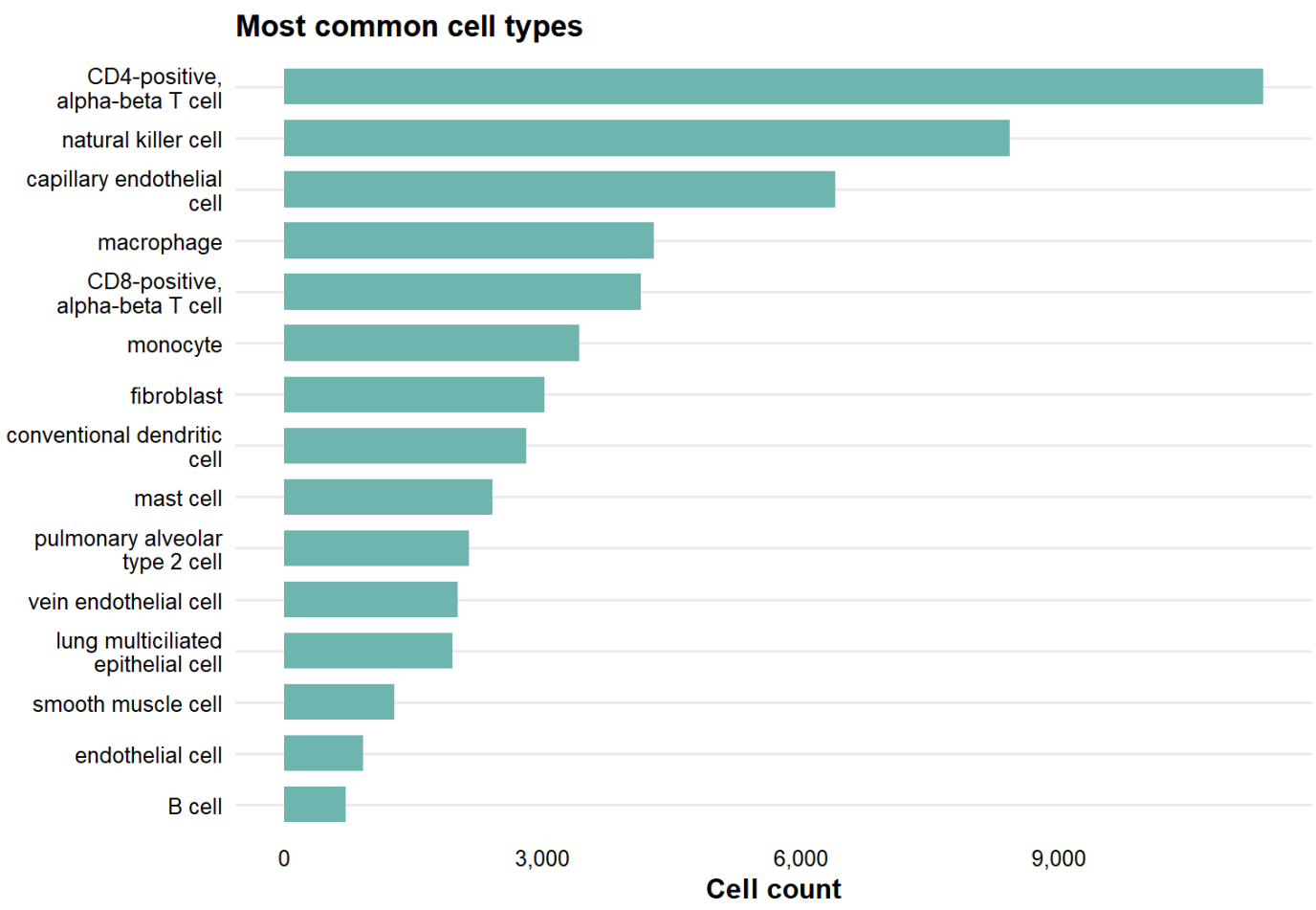
Top 15 cell types by count

cell_type	n
CD4-positive, alpha-beta T cell	11365
natural killer cell	8423
capillary endothelial cell	6393
macrophage	4289
CD8-positive, alpha-beta T cell	4136
monocyte	3425
fibroblast	3022
conventional dendritic cell	2808
mast cell	2413
pulmonary alveolar type 2 cell	2144

cell_type	n
vein endothelial cell	2015
lung multiciliated epithelial cell	1950
smooth muscle cell	1279
endothelial cell	912
B cell	712

The following figure displays the 15 most common cell types by frequency.

```
cell_type_counts %>%
  slice_max(order_by = n, n = 15) %>%
  mutate(cell_type = fct_reorder(cell_type, n)) %>%
  ggplot(aes(x = cell_type, y = n)) +
  geom_col(width = 0.7, fill = "#72B7B2") +
  coord_flip() +
  scale_y_continuous(labels = comma) +
  scale_x_discrete(labels = wrap_axis) +
  labs(
    title = "Most common cell types",
    x = NULL,
    y = "Cell count"
  )
```



Top 15 cell types by frequency in the analytic dataset.

The following code summarizes the number of cells contributed by each recorded sex category.

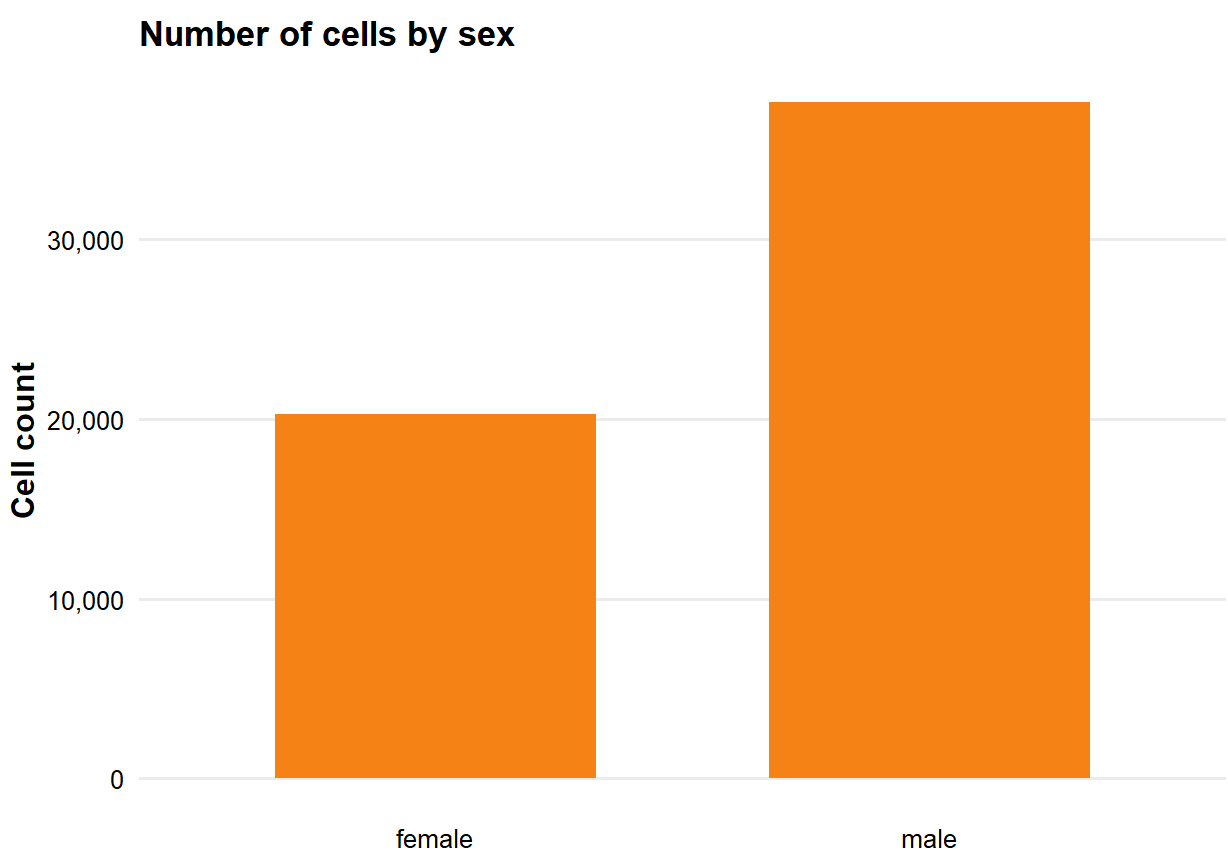
```
sex_counts <- full_df %>%
  count(sex, sort = TRUE)

kable_style(sex_counts, caption = "Cell counts by sex")
```

Cell counts by sex

sex	n
male	37631
female	20287

The following figure shows the distribution of cells by recorded sex.

```
ggplot(sex_counts, aes(x = fct_reorder(sex, n), y = n)) +
  geom_col(width = 0.65, fill = "#F58518") +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Number of cells by sex",
    x = NULL,
    y = "Cell count"
  )
```



Cell counts by recorded sex.

Distribution of key gene expression variables

The following code calculates summary statistics for the selected gene expression variables across all cells.

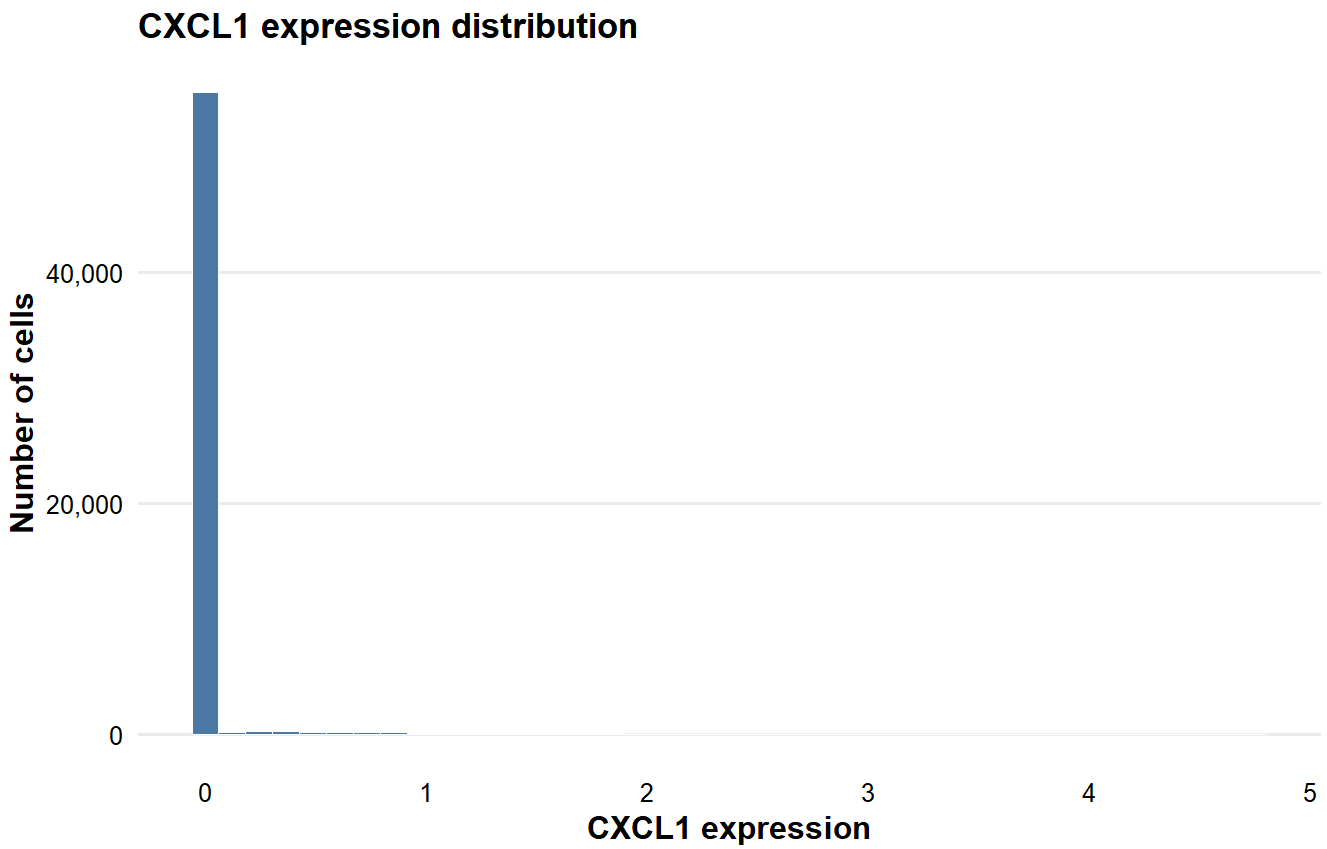
```
gene_summary <- full_df %>%
  summarise(
    CXCL1_mean = mean(CXCL1, na.rm = TRUE),
    CXCL1_median = median(CXCL1, na.rm = TRUE),
    CXCL8_mean = mean(CXCL8, na.rm = TRUE),
    CXCL8_median = median(CXCL8, na.rm = TRUE),
    SFTPC_mean = mean(SFTPC, na.rm = TRUE),
    SFTPC_median = median(SFTPC, na.rm = TRUE),
    AGER_mean = mean(AGER, na.rm = TRUE),
    AGER_median = median(AGER, na.rm = TRUE)
  )

kable_style(gene_summary, caption = "Summary statistics for selected genes")
```

Summary statistics for selected genes

CXCL1_mean	CXCL1_median	CXCL8_mean	CXCL8_median	SFTPC_mean	SFTPC_median	AGER_mean	AGER_media
0.042	0	0.269	0	0.895	0.564	0.047	

The following figure shows the overall distribution of CXCL1 expression across all cells.

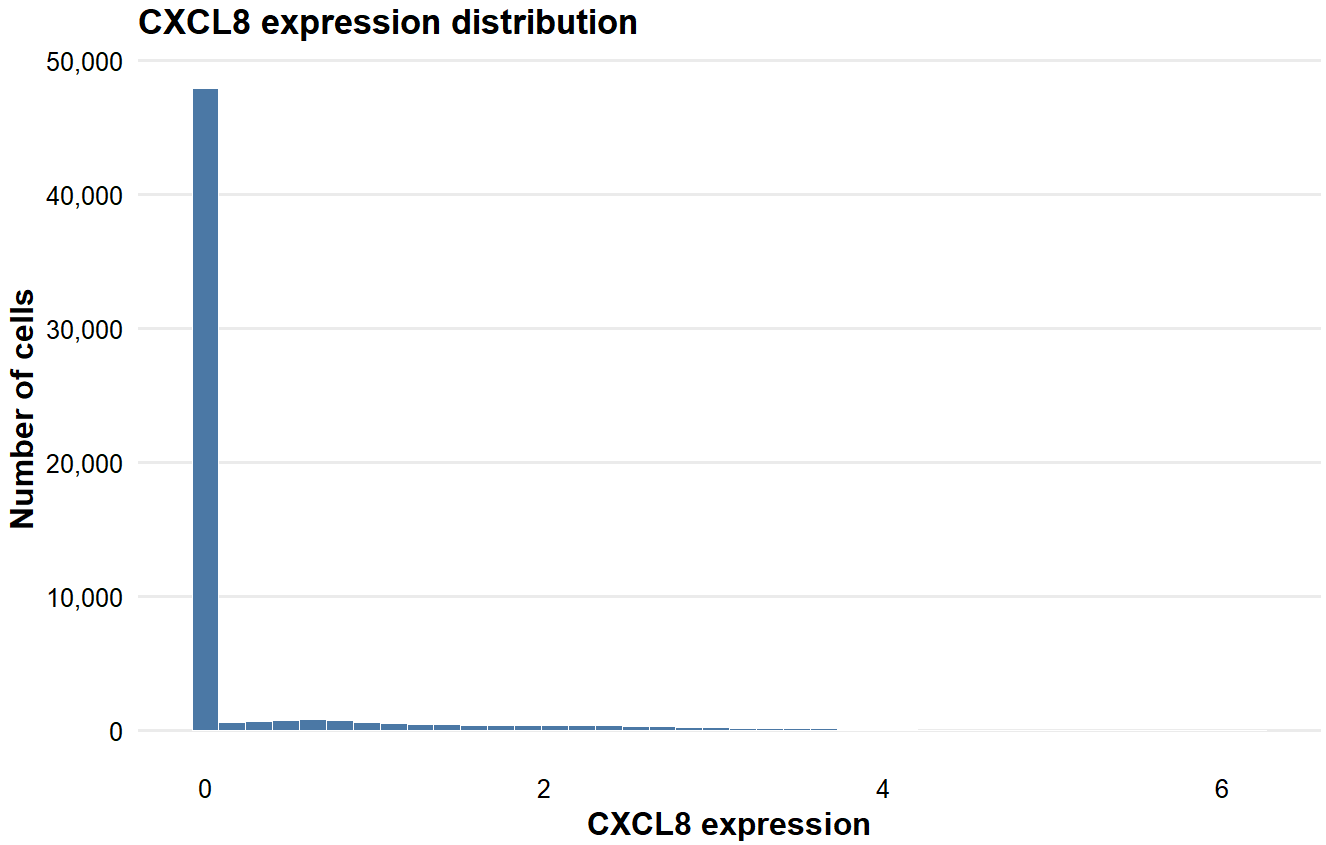
```
ggplot(full_df, aes(x = CXCL1)) +
  geom_histogram(bins = 40, fill = "#4C78A8", color = "white", linewidth = 0.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "CXCL1 expression distribution",
    x = "CXCL1 expression",
    y = "Number of cells"
  )
```



Distribution of CXCL1 expression across all cells.

The following figure shows the overall distribution of CXCL8 expression across all cells.

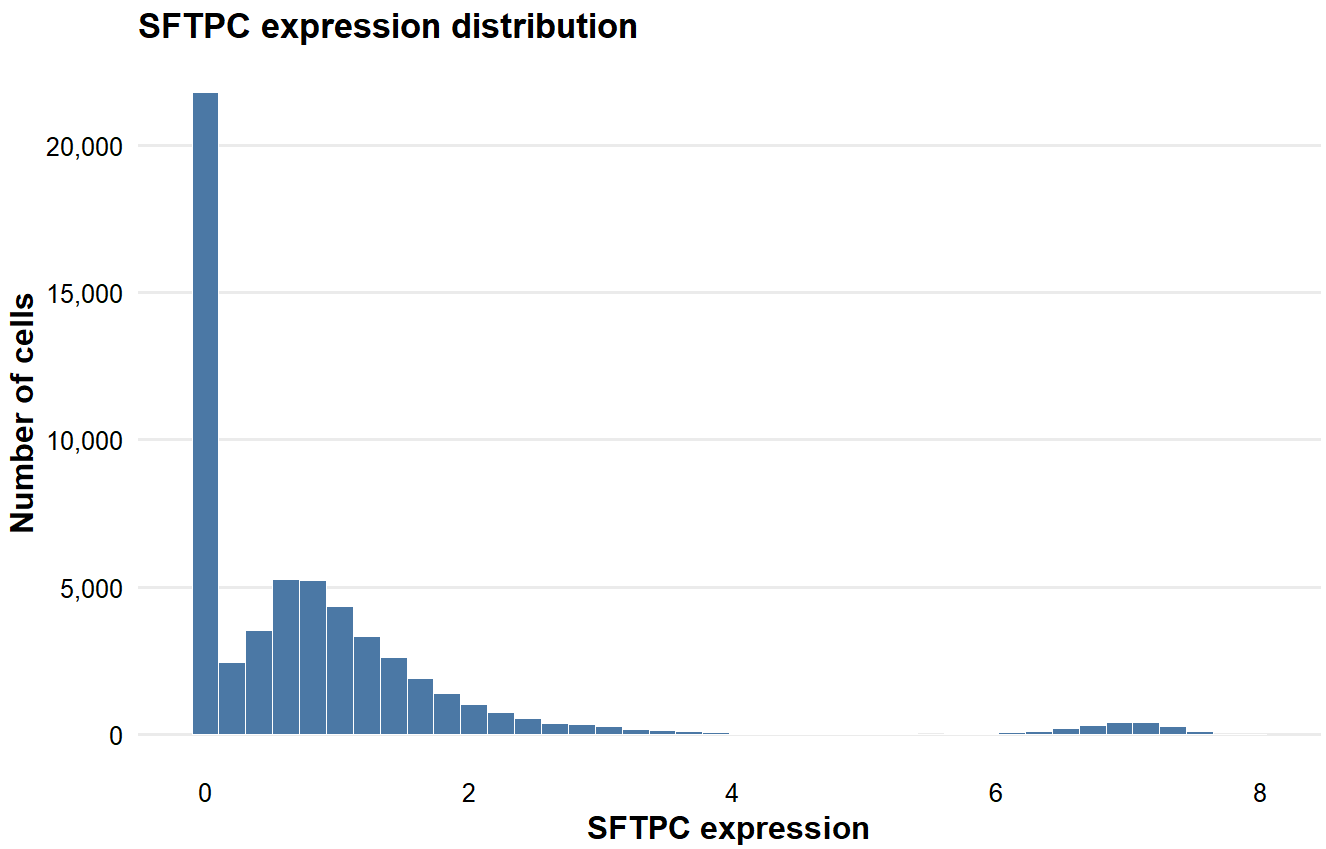
```
ggplot(full_df, aes(x = CXCL8)) +
  geom_histogram(bins = 40, fill = "#4C78A8", color = "white", linewidth = 0.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "CXCL8 expression distribution",
    x = "CXCL8 expression",
    y = "Number of cells"
  )
```



Distribution of CXCL8 expression across all cells.

The following figure shows the overall distribution of SFTPC expression across all cells.

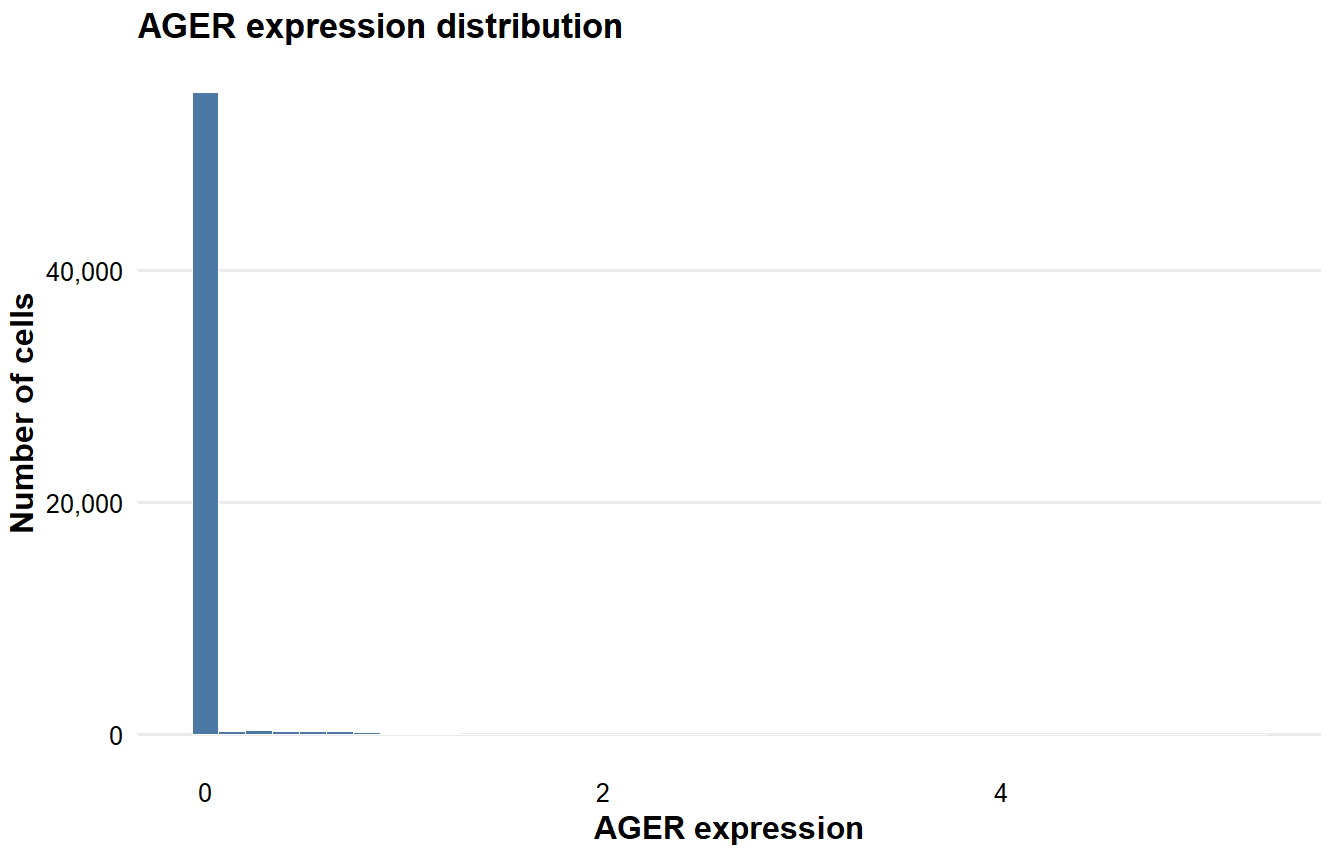
```
ggplot(full_df, aes(x = SFTPC)) +
  geom_histogram(bins = 40, fill = "#4C78A8", color = "white", linewidth = 0.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "SFTPC expression distribution",
    x = "SFTPC expression",
    y = "Number of cells"
  )
```



Distribution of SFTPC expression across all cells.

The following figure shows the overall distribution of AGER expression across all cells.

```
ggplot(full_df, aes(x = AGER)) +
  geom_histogram(bins = 40, fill = "#4C78A8", color = "white", linewidth = 0.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "AGER expression distribution",
    x = "AGER expression",
    y = "Number of cells"
  )
```



Distribution of AGER expression across all cells.

Preliminary Results

Cell type composition across disease groups

The following code summarizes cell type counts and within-disease proportions to evaluate compositional differences across disease groups.

```
disease_celltype_counts <- full_df %>%
  count(disease, cell_type) %>%
  group_by(disease) %>%
  mutate(prop = n / sum(n)) %>%
  ungroup()

kable_style(
  head(disease_celltype_counts, 20),
  caption = "Cell type counts and proportions by disease"
)
```

Cell type counts and proportions by disease

disease	cell_type	n	prop

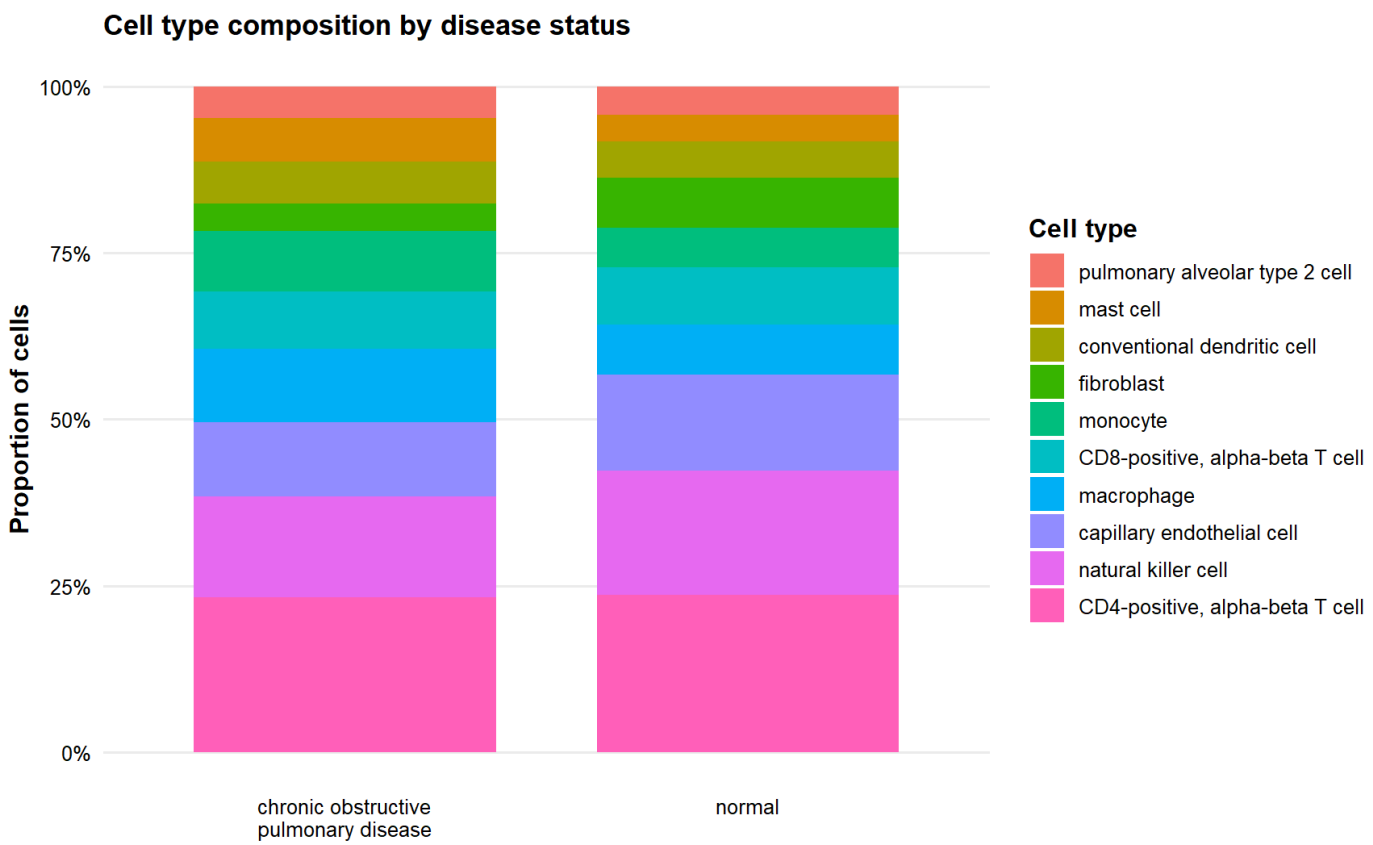
disease	cell_type	n	prop
chronic obstructive pulmonary disease	fibroblast	741	0.034
chronic obstructive pulmonary disease	mast cell	1193	0.054
chronic obstructive pulmonary disease	endothelial cell	325	0.015
chronic obstructive pulmonary disease	club cell	395	0.018
chronic obstructive pulmonary disease	smooth muscle cell	443	0.020
chronic obstructive pulmonary disease	macrophage	2002	0.091
chronic obstructive pulmonary disease	B cell	345	0.016
chronic obstructive pulmonary disease	monocyte	1636	0.074
chronic obstructive pulmonary disease	natural killer cell	2744	0.124
chronic obstructive pulmonary disease	CD4-positive, alpha-beta T cell	4187	0.189
chronic obstructive pulmonary disease	CD8-positive, alpha-beta T cell	1537	0.069
chronic obstructive pulmonary disease	leukocyte	185	0.008
chronic obstructive pulmonary disease	neutrophil	164	0.007
chronic obstructive pulmonary disease	plasma cell	179	0.008
chronic obstructive pulmonary disease	conventional dendritic cell	1132	0.051
chronic obstructive pulmonary disease	pulmonary alveolar type 1 cell	219	0.010
chronic obstructive pulmonary disease	pulmonary alveolar type 2 cell	848	0.038
chronic obstructive pulmonary disease	endothelial cell of lymphatic vessel	127	0.006
chronic obstructive pulmonary disease	capillary endothelial cell	1992	0.090
chronic obstructive pulmonary disease	vein endothelial cell	829	0.037

The following code identifies the most abundant cell types overall and prepares a disease-specific composition dataset for visualization.

```
top_cell_types <- disease_celltype_counts %>%
  group_by(cell_type) %>%
  summarise(total = sum(n), .groups = "drop") %>%
  slice_max(order_by = total, n = 10) %>%
  pull(cell_type)

plot_celltype_comp <- full_df %>%
  filter(cell_type %in% top_cell_types) %>%
  count(disease, cell_type) %>%
  group_by(disease) %>%
  mutate(prop = n / sum(n)) %>%
  ungroup() %>%
  mutate(cell_type = fct_reorder(cell_type, n, .fun = sum))
```

The following figure compares the relative composition of the most abundant cell types across disease groups.

```
ggplot(plot_celltype_comp, aes(x = disease, y = prop, fill = cell_type)) +
  geom_col(width = 0.75) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  scale_x_discrete(labels = wrap_axis) +
  labs(
    title = "Cell type composition by disease status",
    x = NULL,
    y = "Proportion of cells",
    fill = "Cell type"
  ) +
  guides(fill = guide_legend(ncol = 1))
```



Relative cell type composition within each disease group, restricted to the 10 most abundant cell types overall.

Gene expression by disease

The following code computes disease-specific summary statistics for each selected gene to address the primary research question.

```
gene_disease_summary <- full_df %>%
  group_by(disease) %>%
  summarise(
    n = n(),
    CXCL1_mean = mean(CXCL1, na.rm = TRUE),
    CXCL1_median = median(CXCL1, na.rm = TRUE),
```

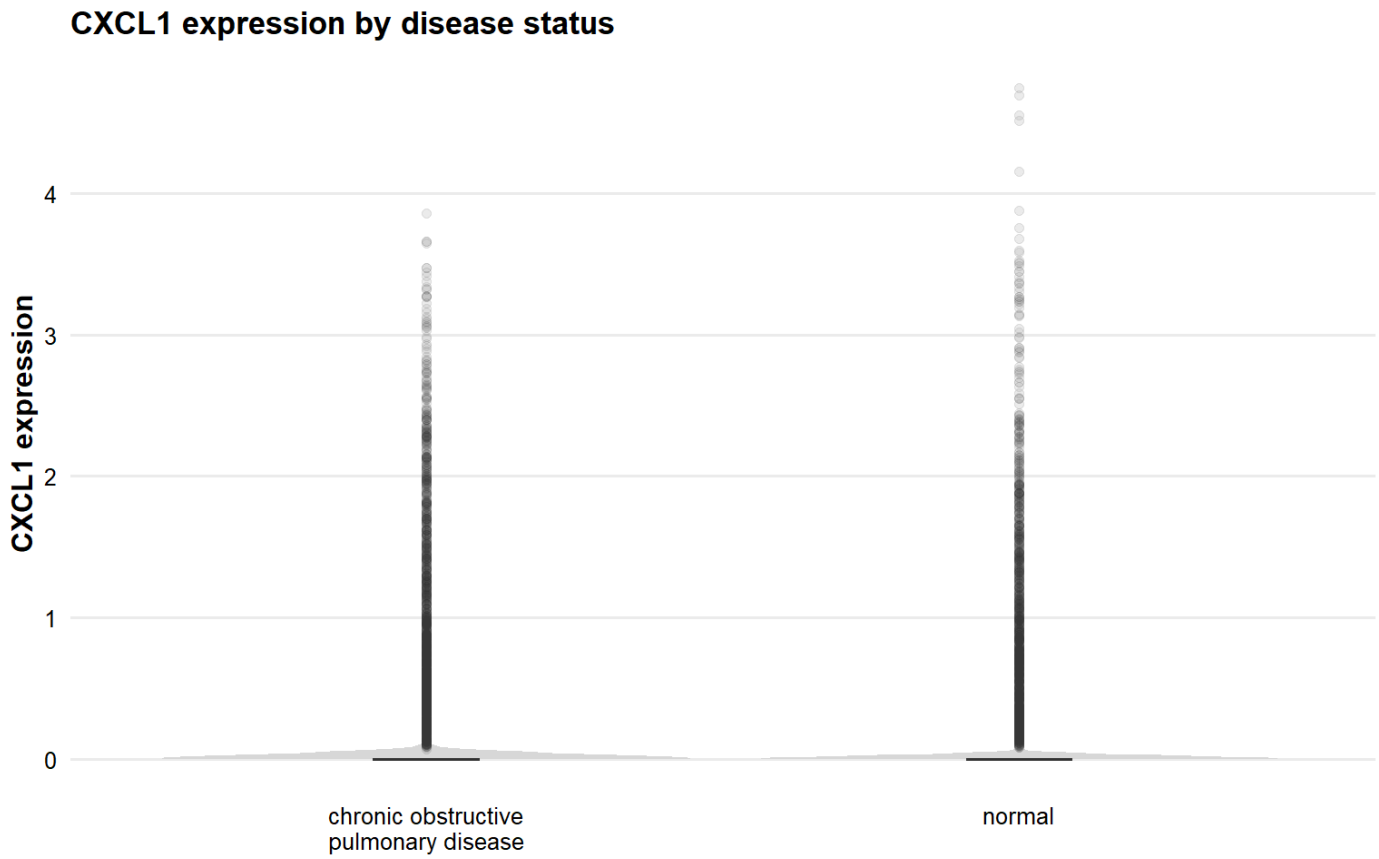
```
    CXCL8_mean = mean(CXCL8, na.rm = TRUE),
    CXCL8_median = median(CXCL8, na.rm = TRUE),
    SFTPC_mean = mean(SFTPC, na.rm = TRUE),
    SFTPC_median = median(SFTPC, na.rm = TRUE),
    AGER_mean = mean(AGER, na.rm = TRUE),
    AGER_median = median(AGER, na.rm = TRUE)
  )

kable_style(gene_disease_summary, caption = "Summary statistics for selected genes by disease sta
```

Summary statistics for selected genes by disease status

disease	n	CXCL1_mean	CXCL1_median	CXCL8_mean	CXCL8_median	SFTPC_mean	SFTPC_median	AGER_r
chronic obstructive pulmonary disease	22119	0.060	0	0.351	0	0.733	0.197	(
normal	35799	0.031	0	0.218	0	0.995	0.705	(

The following figure compares CXCL1 expression distributions across disease groups.

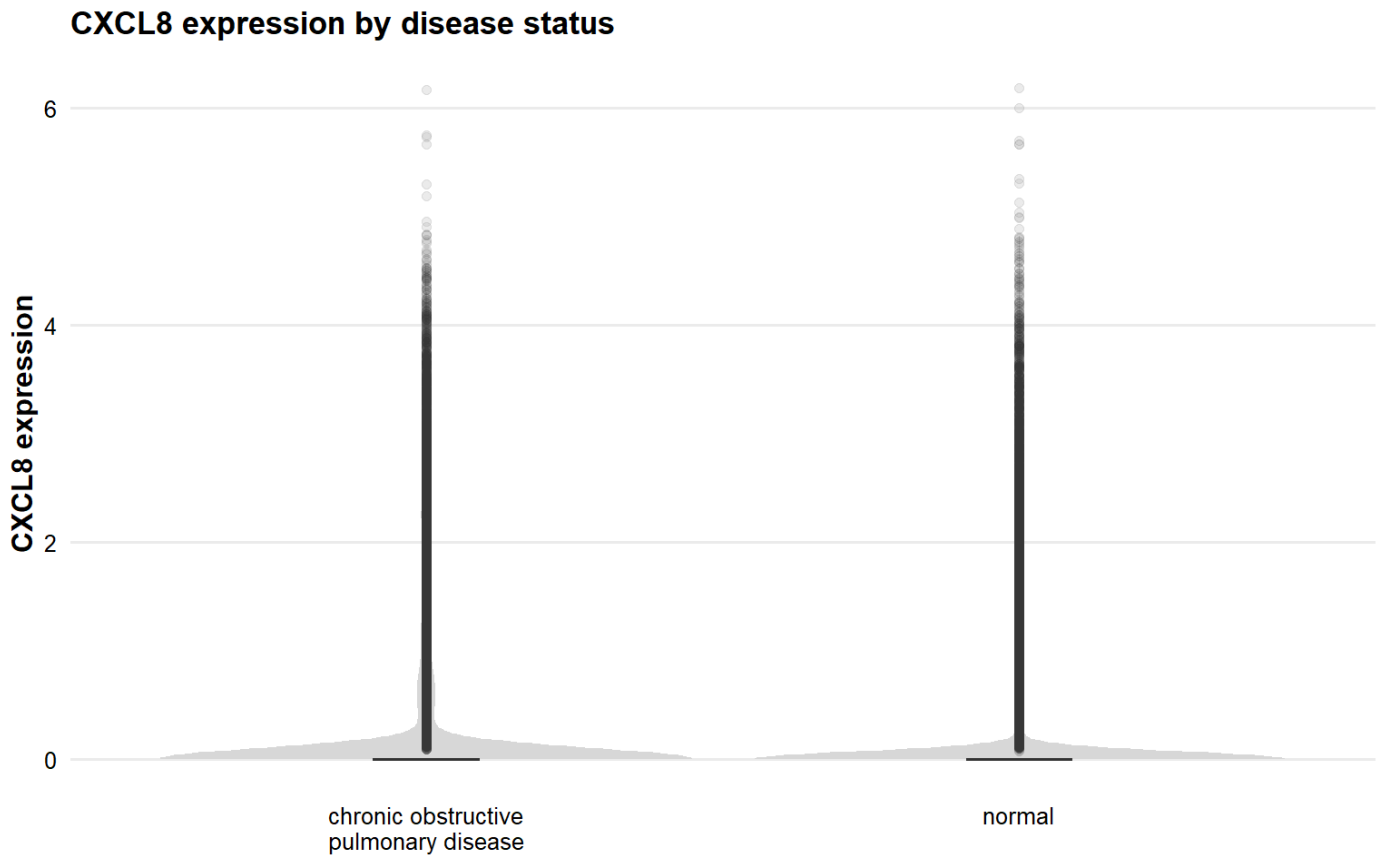
```
ggplot(full_df, aes(x = disease, y = CXCL1)) +
  geom_violin(fill = "gray85", color = NA, scale = "width") +
  geom_boxplot(width = 0.18, outlier.alpha = 0.08, linewidth = 0.3) +
  scale_x_discrete(labels = wrap_axis) +
  labs(
    title = "CXCL1 expression by disease status",
    x = NULL,
    y = "CXCL1 expression"
  )
```



Distribution of CXCL1 expression by disease group. Violin plots show the full distribution, while boxplots summarize the median and interquartile range.

The following figure compares CXCL8 expression distributions across disease groups.

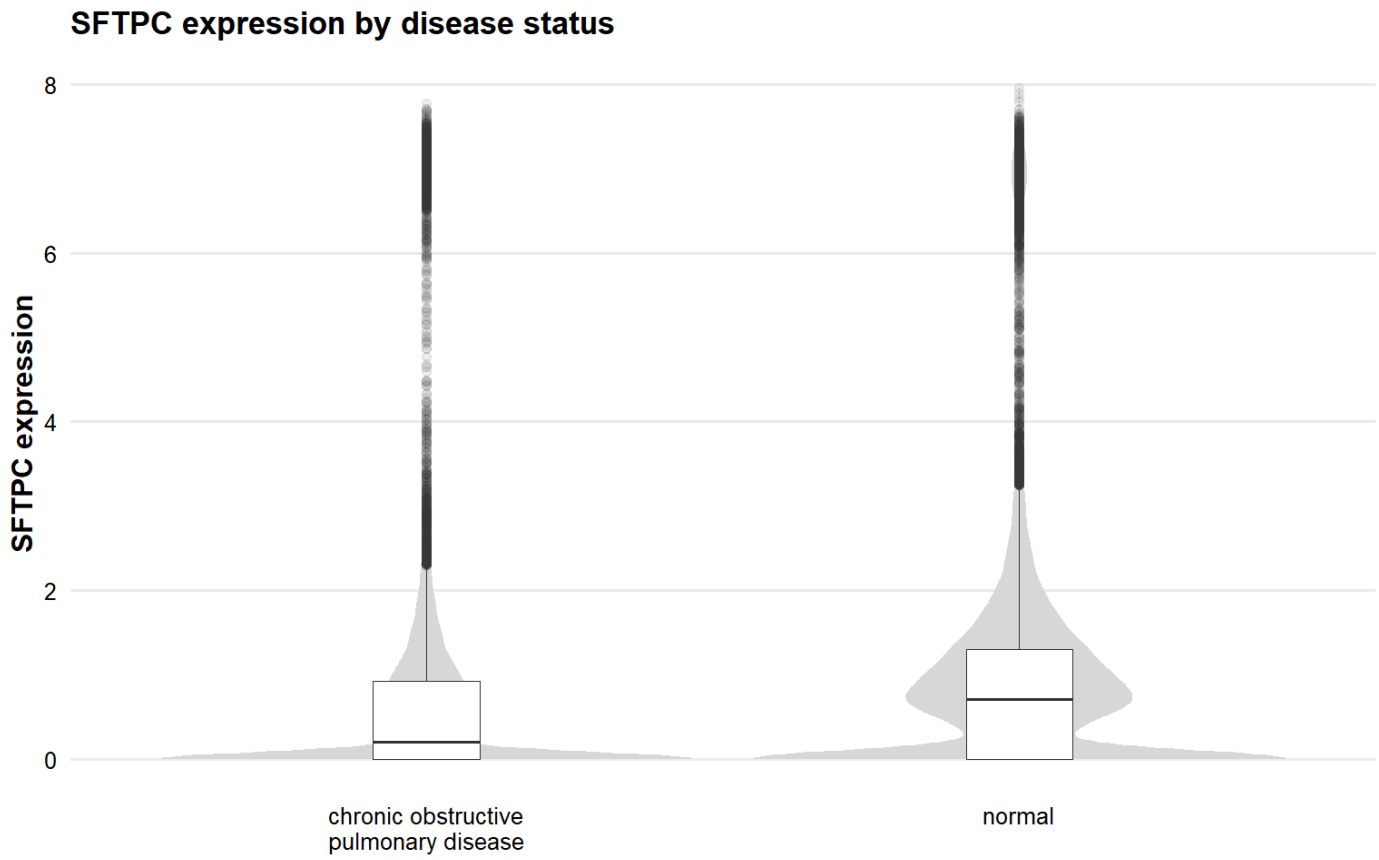
```
ggplot(full_df, aes(x = disease, y = CXCL8)) +
  geom_violin(fill = "gray85", color = NA, scale = "width") +
  geom_boxplot(width = 0.18, outlier.alpha = 0.08, linewidth = 0.3) +
  scale_x_discrete(labels = wrap_axis) +
  labs(
    title = "CXCL8 expression by disease status",
    x = NULL,
    y = "CXCL8 expression"
  )
```



Distribution of CXCL8 expression by disease group.

The following figure compares SFTPC expression distributions across disease groups.

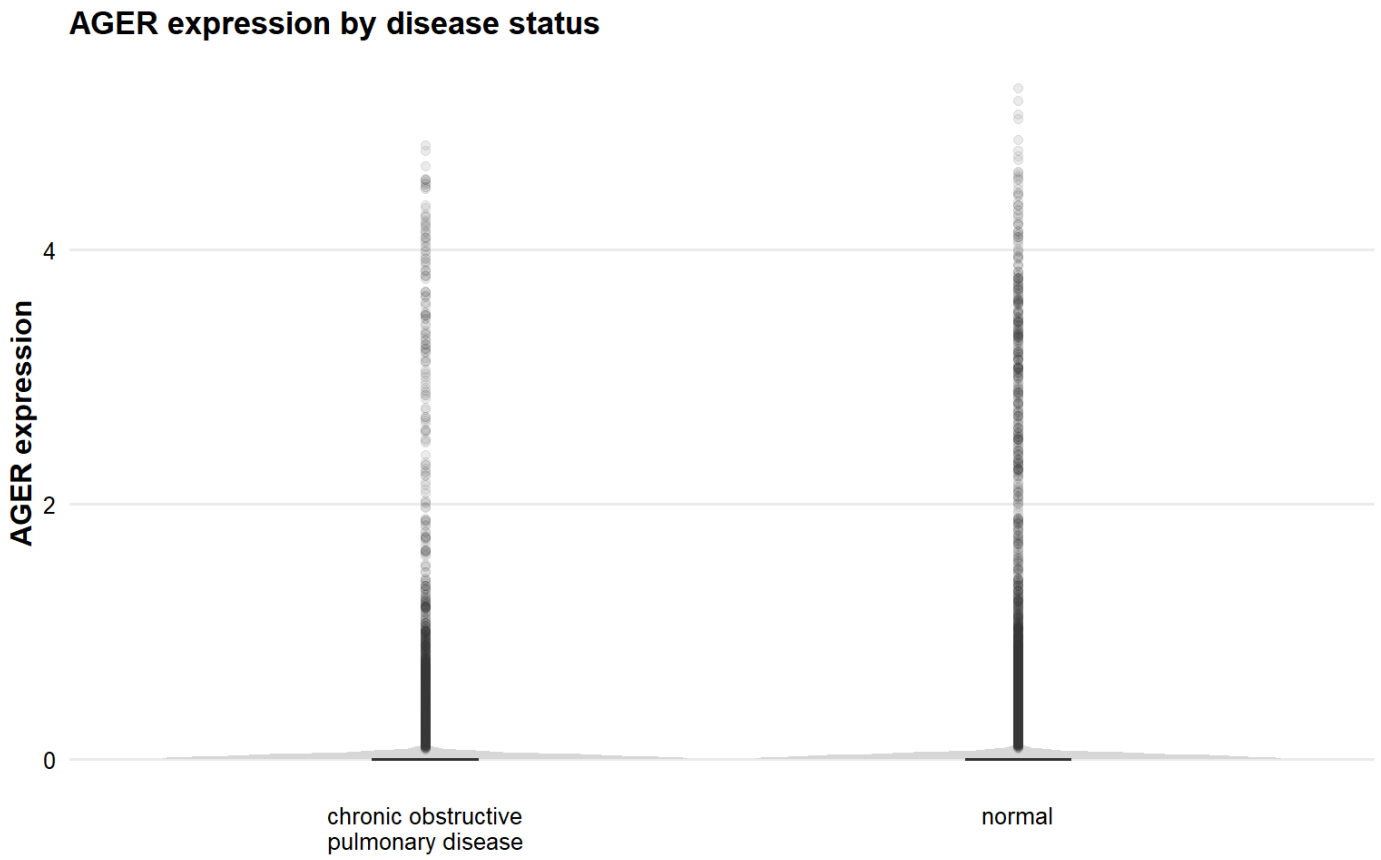
```
ggplot(full_df, aes(x = disease, y = SFTPC)) +
  geom_violin(fill = "gray85", color = NA, scale = "width") +
  geom_boxplot(width = 0.18, outlier.alpha = 0.08, linewidth = 0.3) +
  scale_x_discrete(labels = wrap_axis) +
  labs(
    title = "SFTPC expression by disease status",
    x = NULL,
    y = "SFTPC expression"
  )
```



Distribution of SFTPC expression by disease group.

The following figure compares AGER expression distributions across disease groups.

```
ggplot(full_df, aes(x = disease, y = AGER)) +
  geom_violin(fill = "gray85", color = NA, scale = "width") +
  geom_boxplot(width = 0.18, outlier.alpha = 0.08, linewidth = 0.3) +
  scale_x_discrete(labels = wrap_axis) +
  labs(
    title = "AGER expression by disease status",
    x = NULL,
    y = "AGER expression"
  )
```



Distribution of AGER expression by disease group.

Follow-up analysis: gene expression by disease within cell type

The follow-up analysis focuses on pulmonary alveolar type 2 cells because SFTPC is a canonical marker for this cell population and is especially interpretable within that biological context.

The following code summarizes SFTPC expression within pulmonary alveolar type 2 cells to examine whether disease-related patterns persist in a more homogeneous cell population.

```
full_df %>%
  filter(cell_type == "pulmonary alveolar type 2 cell") %>%
  group_by(disease) %>%
  summarise(
    n = n(),
    SFTPC_mean = mean(SFTPC, na.rm = TRUE),
    SFTPC_median = median(SFTPC, na.rm = TRUE)
  ) %>%
  kable_style(caption = "SFTPC summary statistics in pulmonary alveolar type 2 cells by disease")
```

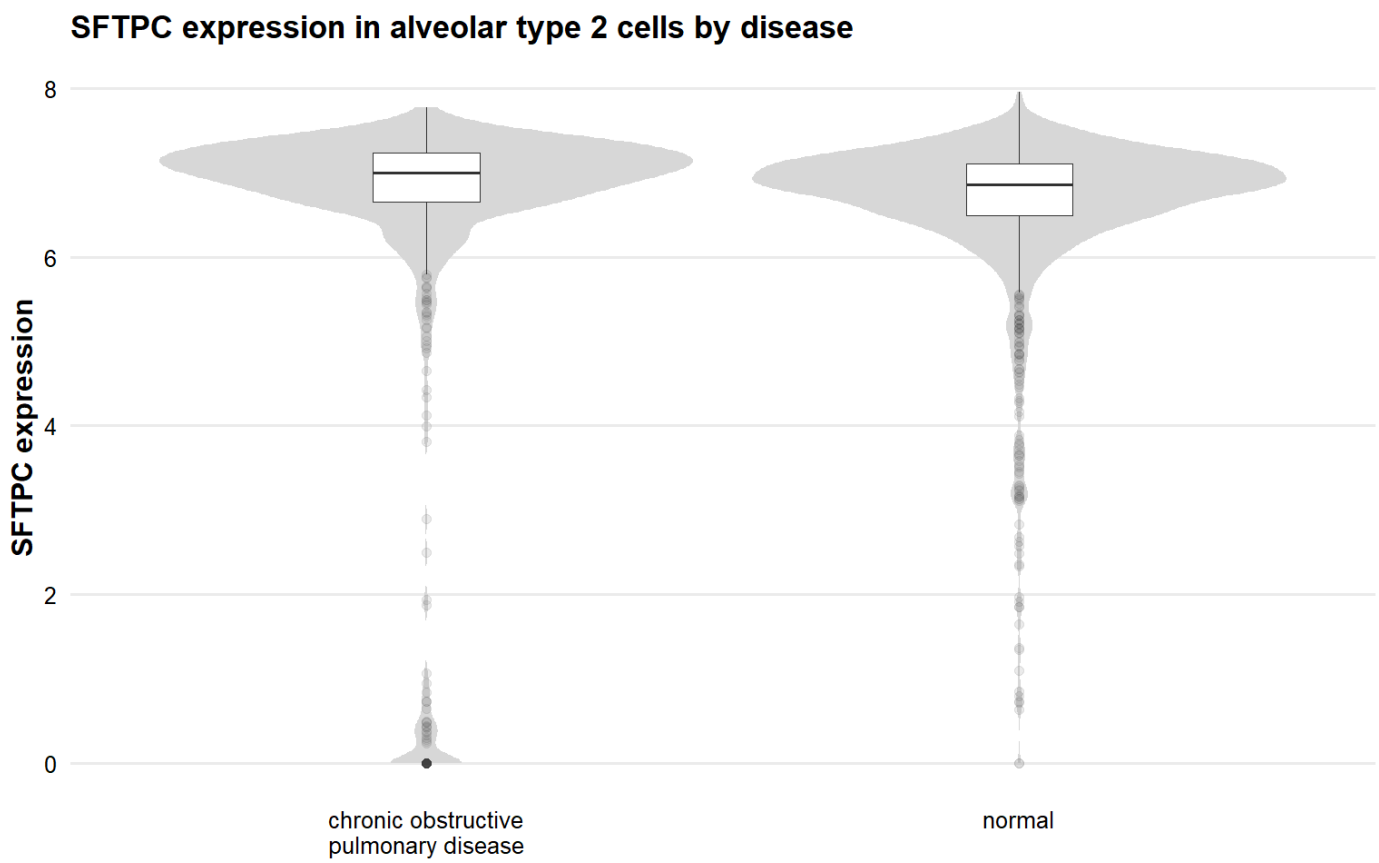
SFTPC summary statistics in pulmonary alveolar type 2 cells by disease

disease	n	SFTPC_mean	SFTPC_median
chronic obstructive pulmonary disease	848	6.513	6.996

disease	n	SFTPC_mean	SFTPC_median
normal	1296	6.610	6.852

The following figure compares SFTPC expression across disease groups within pulmonary alveolar type 2 cells.

```
full_df %>%
  filter(cell_type == "pulmonary alveolar type 2 cell") %>%
  ggplot(aes(x = disease, y = SFTPC)) +
  geom_violin(fill = "gray85", color = NA, scale = "width") +
  geom_boxplot(width = 0.18, outlier.alpha = 0.08, linewidth = 0.3) +
  scale_x_discrete(labels = wrap_axis) +
  labs(
    title = "SFTPC expression in alveolar type 2 cells by disease",
    x = NULL,
    y = "SFTPC expression"
  )
```



SFTPC expression across disease groups within pulmonary alveolar type 2 cells.

Conclusion

Overall, the rendered results suggest meaningful differences between normal and chronic obstructive pulmonary disease (COPD) samples in both cell composition and gene expression. The dataset included

57,918 cells from 16 donors and 16 samples, with more cells from normal tissue than COPD tissue (35,799 vs. 22,119), and no missing data in the variables analyzed. The cell-type composition plots indicate that the relative abundance of major immune and epithelial populations differs by disease group, supporting the idea that COPD is associated with shifts in the cellular makeup of lung tissue. Gene-level summaries also showed disease-related expression differences: compared with normal samples, COPD samples had higher mean expression of CXCL1 (0.060 vs. 0.031) and CXCL8 (0.351 vs. 0.218), but lower mean expression of SFTPC (0.733 vs. 0.995) and AGER (0.040 vs. 0.051). In the follow-up analysis restricted to pulmonary alveolar type 2 cells, mean SFTPC expression was still slightly lower in COPD than in normal cells (6.513 vs. 6.610), suggesting that the disease-associated reduction in SFTPC is not explained only by broad differences in cell-type composition. Together, these findings provide preliminary evidence that COPD is associated with both altered lung cell populations and shifts in expression of biologically relevant marker genes.

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